

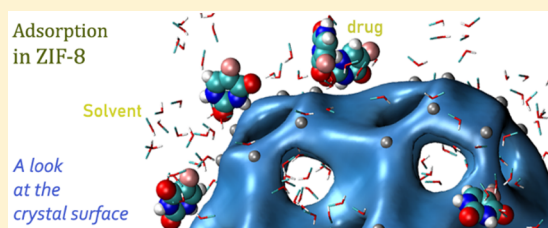
Simulation of the Adsorption and Release of Large Drugs by ZIF-8

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S Supporting Information

ABSTRACT: The adsorption and release of two drugs SFU (5-fluorouracil) and CAF (caffeine) into and from the ZIF-8 framework were simulated by the Gibbs-ensemble Monte Carlo approach employing two models for representing the sorbent: one without surface (ZIF-8P) and another with surface (ZIF-8S). The inner pores of ZIF-8S were inaccessible to the drugs, but accessible to the solvents (methanol or water). The ZIF-8P model is not recommended to describe the actual sorption processes because it lacks surface and solvent effects, which are reflected in the poor quantitative agreement with experimental results. The ZIF-8S model yielded results for the sorption of CAF in very close agreement with the experimental loading from methanol solution and release of the drug into water. For SFU, the computer simulations provided qualitative agreements, which suggests that the sorbent–SFU interaction potentials should be improved. The excellent performance of the ZIF-8S model is due to its adequate description of the surface and by exposing adsorption sites such as undercoordinated zinc ions to interactions with large molecules. This was achieved by applying periodic conditions to a ZIF-8 nanocrystal, instead of an elementary cell, which is easy to generalize and used to describe several surface defects. Furthermore, the combination of this ZIF-8S model with the Monte Carlo method provides a very simple and efficient approach to simulate the inaccessibility of the ZIF-8 inner porosity to large molecules. Namely, any trial moves that inserted the drug within the pore were disregarded. This is a quite simple and general approach that can be promptly applied to a large number of MOF sorbents and of drugs that cannot access the inner pores.



INTRODUCTION

Computer simulations of adsorption and desorption processes are relevant, especially for drug delivery systems, because they can provide guidelines and aid the design and development of new and/or improved adsorbing materials.^{1–3} Performing these computer simulations require the representation of the real system and of the processes, denoted as chemical model, as well as the interaction potentials.⁴ These latter requirements are being developed and improved for many decades, which provide a wide range of choices regarding accuracy, transferability, specificity, dependence on the system, and computational speed, among others. Meanwhile, investigations regarding the chemical model are more scarce and some representations are not feasible for practical applications or are not adequate to describe the actual system and/or process of interest. Thus, one of the main concerns of this contribution regards the choice of the chemical model to simulate the adsorption and desorption processes of model drugs on/from porous materials.

One of the most promising materials for these processes are metal–organic frameworks (MOFs) because of their tunable and controllable properties,^{5–7} which are finding a wide range of potential and actual applications in industry, environment, and healthcare as well as in basic research.^{2,5,8–10} One of these properties of interest in adsorption/desorption are their large free volume and surface area that provide their high permanent porosity.^{2,5} Combining these properties with tunable chemical environments allows the interaction, uptake, and retention of

molecules within the framework and on their surfaces, making MOFs suitable materials for adsorption, storage, and controlled release of organic compounds.^{11–14}

Among the great variety of MOFs, zeolitic imidazolate frameworks (ZIFs) have shown versatility for several applications, especially in biomedicine.^{15–19} For instance, ZIF-8 is a well-characterized crystalline material based on 2-methyl imidazolate linkers tetrahedrally coordinated to zinc ions, forming large cavities of 11.6 Å of diameter with small opening-windows of 3.4 Å between the cages.²⁰ This structure provides intrinsic high-encapsulation and slow-release properties to this framework,^{20–22} which were used, for instance, in drug-delivery systems in anticancer therapy,^{21–24} drug carriers of bioactive molecules,^{17,25,26} and stimuli-responsive nanomaterials for controlled drug release,^{13,14,27} to mention a few. Notice, however, that ZIF-8 has a low-diffusivity and slow-release of large drugs, which then relates the adsorption/desorption processes to surface phenomena instead of porous/cavity ones.¹⁴ In this regard, computer simulations and molecular modeling were important to show that the diffusion of 5-fluorouracil (SFU) and caffeine (CAF), model drugs, between neighboring pores in ZIF-8 is severely restricted due

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to large energy barriers.¹⁴ In fact, these observations can explain the small loadings and rapid or poorly controlled release of an important anticancer drug doxorubicin, because it is too large to enter the pores of ZIF-8.^{23,26} As a result, several approaches to incorporate large drugs into ZIF-8 and other MOFs employ in situ procedures that encapsulate the drugs during the synthesis.^{2,21,26} These approaches can become more expensive and difficult to perform, which decrease their viability for biomedical applications.⁵ Indeed, these findings suggest that to improve the performance of ZIF-8 and other MOFs for upload/release of large molecules the focus should be in their surfaces and the outermost free cavities, instead of their inner pores and channels.

In this context, ZIF-8 is a suitable system because its surface and other relevant issues related to diffusion of large molecules in confined spaces have been previously investigated from both experimental and computational points of view.^{28–31} For instance, point defects such as the vacancies of imidazolate linkers or of metal centers in ZIF-8 have been reported.^{31–33} It has also been shown that extended structural defects, large external void-cavities and impurities on the surfaces of ZIFs have direct effects on their reactivity with guest molecules.²⁸ However, these reported simulations and calculations were performed under periodical conditions of the structure and the chemical models had no surfaces.^{31–33} Therefore, the inherent surface defects due to the finite size and shape of the nanocrystals in solution would not appear in such chemical models. As a consequence, the adsorption/desorption processes of large molecules by ZIF-8 are not fully understood, especially from a computational point of view, which requires the inclusion of the solvent effects and of the surface defects to properly represent the actual material. Thus, combining an adequate description of the surface with the solvent, particularly, water because of the composition of the physiological media⁸ and for being the preferred solvent in delivery experiments with MOFs,^{1,14} seems essential to simulate these adsorption/desorption processes.

Therefore, the effects of how to represent the actual system and the processes are the main concern of this work. To perform such analysis, we have chosen to describe the adsorption and release of 5FU and CAF (drugs) at and from the ZIF-8 framework (host) in solution (water and methanol), because some previous computer simulations are available, in addition to several experimental results and structural characterizations. The computational modeling and simulations were performed at the atomistic level, with explicit solvent molecules and considering the surface and defects on the framework.

■ COMPUTATIONAL PROCEDURES

Monte Carlo (MC) computational simulations based on the approaches^{30,34} implemented in the RASPA software package were performed.³⁵ The adsorption was considered for pure drugs and pure solvents (methanol and water) as well as for solutions of the drugs in methanol, whereas the drug release was simulated by the ZIF-8 loaded with the drug and free of solvent immersed into water. The role of the surface on these processes was assessed by performing simulations with two chemical models: (i) a surface-free model of ZIF-8, denoted as ZIF-8P; and (ii) a surface model, named ZIF-8S.

Molecular Models. The molecular structures of the drugs 5FU (5-fluorouracil) and CAF (caffeine), solvents (methanol and water), and of the building blocks of the ZIF-8 framework

are illustrated in Figure S1, which also shows the atomic labels employed in the intermolecular potential and models.

The pristine ZIF-8 unit cell (CSD reference no. 602542)²⁰ was used to construct the ZIF-8P model (see details in Table S1 and Figure S2), consisting in a $2 \times 2 \times 2$ ZIF-8 cubic cell of 33.982 Å with a BET surface area of $1623.1 \pm 0.2 \text{ m}^2 \text{ g}^{-1}$. An expansion model was used to construct the ZIF-8S structure by removing the outermost imidazolate linkers on the surface of a $4 \times 4 \times 6$ ZIF-8P cubic cell, yielding a 61.964 Å cubic cell (Figure S3) with a BET surface area of $1843.4 \pm 0.2 \text{ m}^2 \text{ g}^{-1}$. This procedure preserves the internal porosity of ZIF-8P and exposes half void cavities onto the surface (details in Figures S3 and S4). It is noteworthy that the available surface area of the ZIF-8S model is in very good agreement to the crystallographic ZIF-8 structure ($1947 \text{ m}^2 \text{ g}^{-1}$), which emphasizes the reliability of the ZIF-8S model, which considers only the external surface of the nanocrystal, whereas the ZIF-8P model includes the full inner-porosity. The CIFs of the ZIF-8P and ZIF-8S models are available as Supporting Information. Because the flexibility of ZIF-8 was shown to have negligible effects on its surface properties,^{31,32,36} and no appreciable variation at physiological conditions (25 °C and 1 atm),³⁷ both ZIF-8 models were treated as rigid, as well as the drugs and water. Methanol was treated as semiflexible by allowing the COH angular bending.

The methyl groups ($-\text{CH}_3$) in all structures were considered as united atoms, weighing 15.03452 atomic units and labeled as Cm in methanol; C6m, C7m, and C8m in CAF; and C3m in both ZIF-8 models (Figure S1).

Intermolecular interactions were described by 12–6 Lennard-Jones (L-J) plus Coulomb potentials. The L-J parameters for 5FU, CAF, and ZIF-8 were taken from the generic-MOF force field in the RASPA code,³⁵ with refinement from a recent force field for adsorption in MOFs with unsaturated centers.³⁸ The OPLS/2016³⁹ and TIP3P⁴⁰ models were used for methanol and water, respectively, but with L-J interactions added to the hydrogen atoms of water,⁴¹ which improves the description of hydrogen bonds.⁴¹ The atomic charges for both ZIF-8 models were obtained via the charge equilibration method in RASPA,⁴² while the CHELPG//B2GPPLYP/6-31+G(d,p) method⁴³ was used for the drugs. The partial charges of the solvents were kept the same of the respective reference models. The Lorentz–Berthelot mixing rules were used to determine the L-J parameters for cross-interactions and all potentials were shifted to zero at a radial cutoff of 12 Å, with no tail-correction. All these parameters along with the atomic charges to model the electrostatic interactions are detailed in Table S2.

Computational Simulations. The adsorption and release of the drugs in solution were studied via Gibbs-ensemble MC simulations (GEMC),⁴⁴ where two separate simulation boxes are used: one for the adsorbing phase containing the ZIF-8 models, hereafter named S_0 , and the other, denoted as S_1 , for the fluid phase containing the drug in methanol or water. The S_0 boxes were the ZIF-8P and ZIF-8S rigid cubic cells and a $50 \times 50 \times 50$ (Å) cubic box was used for S_1 . Periodic boundary conditions were applied to S_0 and S_1 , and the Ewald summation method was used to calculate the long-range electrostatic interactions.⁴⁵ The inner pores of ZIF-8S were restricted for the drugs and allowed to solvents (Figure S4 for details) in order to mimic the situation that large molecules (drugs) cannot be inserted into the cavities of this framework. The whole system ($S_0 + S_1$) was held at constant pressure,

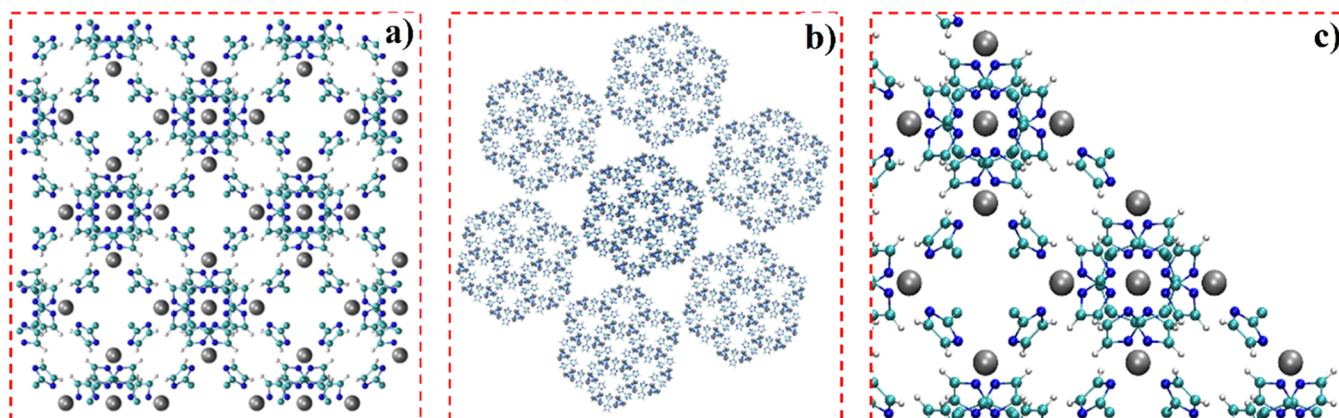


Figure 1. Illustrations of (a) the computational cell of the ZIF-8P model; (b) the periodic structure of the ZIF-8S model; and (c) a section of the surface of the ZIF-8S model. For more details, see Figures S2, S3, and S4 in the [Supporting Information](#).

temperature and composition, while the number of molecules in the individual boxes was changed until the chemical potentials were equalized.⁴⁶

The adsorption enthalpy was calculated through Equation S1 ([Supporting Information](#)), by using the energy-per-particle fluctuation method to approximate the change in the potential energy of the system (U) upon adsorption during GCMC and GEMC simulations, as implemented in the RASPA code.³⁵ This method was tested first for several loadings in order to ensure that the systems are far enough from the saturation state, so that this approach is valid.

For adsorption, 12 solutions of the drugs (SFU and CAF) in methanol at mole fractions from 0.01 to 0.30 were modeled in S_1 (specific composition in [Table S4](#)). The initial configurations of S_1 were created by randomly placing the corresponding number of drug molecules ([Table S4](#)) in the box and avoiding overlap with the other molecules. Simulations consisted of 10 000 NVT-MC cycles for initialization and then 100 000 GEMC cycles for data collection and statistics.

For estimating the ZIF-8 capacity to release the drugs, the initial configurations of S_0 were obtained from the maximum loading after the adsorption step and by removing all methanol molecules adsorbed into the framework (details of desorption in [Table S7](#)). This procedure mimics the drying process of ZIF-8 after loading, in a typical adsorption/desorption experiment.¹⁴ The bulk fluid phase (S_1) for the desorption process consisted of 899 molecules of water in a $30 \times 30 \times 30$ (Å) cubic box. The initialization and production steps for the desorption were the same as for the adsorption.

For all GEMC simulations, the following MC moves were applied: particle translations, rotations, reinsertions within the individual S_0 and S_1 simulation boxes, and particle transfer between the boxes (Gibbs swap move), which removes a particle from one box and insert it into the other. In addition, to reduce the occurrence of overlapping from the MC insertion moves and to increase the number of successfully insertions, the configurational-bias MC scheme was used, where the large molecules were fractionally grown in the more favorable configuration.³⁴

RESULTS AND DISCUSSION

Before presenting the results, it is instructive to discuss in more detail the choices of the chemical model to represent the sorption processes. The ZIF-8P model employs a $2 \times 2 \times 2$

ZIF-8 unit cell which under periodic conditions has no surface, as depicted in [Figure 1a](#). Therefore, the molecules (drugs and/or solvents) can only adsorb into the pores of the ZIF-8. Regarding the adsorption process, it can occur from a reservoir of the pure substance (drugs or solvents) or from a solution of the drug, where methanol was employed as solvent to conform the experiments. The simulations with the pure drugs have been performed previously in [ref 14](#), whereas 12 simulations were performed for methanol solutions of drug at mole fractions varying from 0.01 to 0.3. The main goals of these simulations with the ZIF-8P model were to ascertain and quantify the effects of the solvent, solution concentration, and of the nature of the drugs upon the adsorption, as well as the lack of surfaces. These latter effects can be assessed by comparing the results obtained by employing the ZIF-8P and ZIF-8S models, because even with periodic conditions, the ZIF-8S model has surfaces. An illustration of this model is depicted in [Figure 1b](#), where a $4 \times 4 \times 6$ ZIF-8 expanded cell was used as a computational cell under periodic conditions. Indeed, this ZIF-8S model should be a more adequate representation of the actual ZIF-8 adsorbent, which is composed of micro- or nano-crystals permanent porosity under the sorption experimental conditions. In addition, the adsorption occurs from a methanol solution of the drugs,¹⁴ so a methanol solution at 0.3 mole fraction of each drug was used in the simulations. Another relevant aspect regarding the adsorption of large molecules by ZIF-8 is the large energy barriers involved in the migration of these molecules between adjacent pores.¹⁴ Thus, for adsorption from a liquid phase, the inner porosity of ZIF-8 should be inaccessible to molecules like SFU and CAF. This restriction should then be taken into account during the simulations, which was indeed considered by skipping any Monte Carlo trial moves that inserted the drugs within inner pores. The comparisons between these simulations should quantify the effects of the chemical representation on the sorption processes.

Because most sorption experiments involving MOFs are performed in solution,⁵ the inclusion of the solvent in the computer simulations is important to improve the representation of the actual processes. Usually, the solvents in these experiments are hydrogen bond forming liquids, e.g. water and alcohols, and their explicit description in the computer simulations is commonly required. The water and methanol models used in this work were ascertained and tested by comparing simulated results with experimental data, such as

vapor–liquid equilibrium (VLE) coexistence curves estimated from GEMC for water and methanol, which are presented in Figure S5 and radial distribution functions (RDF) obtained from NVT-MD simulations shown in Figure S6.

The estimated critical temperature T_c (ca. 508 K) and critical density ρ_c (ca. 270 kg m⁻³) using the OPLS/2016 model³⁹ are in very good agreement with the recommended reference values 512.6 K and 281 kg m⁻³ for methanol.⁴⁷ It is noteworthy that including L-J parameters for the hydrogen atoms in water modeled by the simple TIP3P model (see Table S2), which are not present in more sophisticated model potentials, significantly improves the calculated values of the critical properties, namely $T_c \cong 625$ K and $\rho_c \cong 330$ kg m⁻³ (Figure S5), compared to the experimental values 647.1 K and 322 kg m⁻³, respectively, for water.⁴⁸ Furthermore, the calculated densities from NPT-MC calculations at 298 K and 1 atm were 784.3 ± 10.6 and 1009.1 ± 9.8 kg m⁻³ for methanol and water, respectively, which are in excellent agreement with the experimental densities 787 and 997 kg m⁻³ at these same conditions.⁴⁹ These agreements between calculated and experimental data together with the shape and areas of the peaks in the RDFs of these liquids (see Figure S6)⁴¹ indicate that the model potentials employed for methanol and water are quite suitable for computer simulations of these liquids.

The description of the sorbent material is quite relevant for an adequate modeling of the actual systems. For sorption processes, the pore size distribution (PSD) and the powder X-ray diffraction (PXRD) patterns of the sorbent material are properties that should be well represented and modeled. Indeed, the calculated PSD and PXRD using the ZIF-8P model (Figure S7) reproduce the corresponding available experimental characterization of ZIF-8,²⁰ as well as the estimates in previous simulations.¹⁴ Thus, the ZIF-8P model represents the actual structure of ZIF-8 and its porosity due to the pores with 11.6 Å of diameter, interconnected through windows of 3.4 Å formed by six imidazolate linkers.

These systems were selected for testing the chemical models, because the adsorbent (ZIF-8) is well-characterized as observed above and the sorption of SFU and CAF is also available experimentally for comparison. For instance, the experimental loadings of SFU and CAF onto ZIF-8 from methanol solutions are 25.4 wt % (340 mg g⁻¹) and 12.2 wt % (139 mg g⁻¹), respectively, at room temperature.¹⁴ The experimental desorption into water at room temperature, which slowly increases in time, reaches up to 5.7% and 12.5% for SFU and CAF, respectively.¹⁴

Adsorption of the Pure Drugs and Solvents onto ZIF-8P. The complete data with results from grand canonical MC (GCMC) simulations are presented in Table S3, for the adsorption of the pure drugs SFU and CAF as well as the pure solvents methanol and water into ZIF-8P. These results are summarized and illustrated in Figure 2 for the loadings, relative U_{HA} , and the adsorption enthalpy, H_{ads} , where U_{HA} quantifies the percentage of the host-adsorbate interaction energy in the total internal energy U .

Notice that the results presented in Figure 2 can be considered converged and equilibrated because after 30 000 MC cycles, both U and U_{HA} remain practically constant for all systems as can be observed in Figure S8. The loadings of pure SFU and CAF into ZIF-8P model (calculated previously in ref 14), which has no surface and no solvent molecules, are 477 and 422 mg g⁻¹, respectively (Figure 2a), which clearly

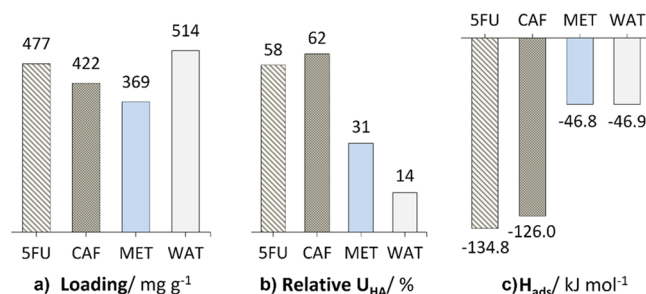


Figure 2. Adsorption of pure components into ZIF-8P simulated by GCMC: (a) Loadings (mg g⁻¹), (b) Relative U_{HA} (%) calculated as $100 \times U_{HA}/U$, and (c) adsorption enthalpy, H_{ads} (kJ mol⁻¹). SFU = 5-fluorouracil, CAF = caffeine, MET = methanol, WAT = water, U_{HA} = host-adsorbate energy, U = total internal energy.

overestimate the experimental adsorption for both drugs (340 and 139 mg g⁻¹), especially for CAF. When comparing (mg g⁻¹), the drug loadings are larger than that for methanol, because of the low density of liquid methanol. However, in terms of the number of adsorbed molecules, the ZIF-8P model absorbs 3 to 4 times more molecules of methanol than that of drugs (see Table S3).

The equilibrated values of U_{HA} are -3.3 , -3.2 , -3.4 , and $-3.6 (\pm 0.1) \times 10^3$ kJ mol⁻¹ (see Table S3) for the systems with pure SFU, CAF, methanol, and water, respectively, which is almost constant within the distinct adsorbates. This indicates that the number of the adsorbing sites within the pores of ZIF-8 is constant as well as their adsorbing strengths regardless the nature, size, or shape of the adsorbate. This is interesting because it indicates some homogeneity of the pores of ZIF-8 and for a given adsorbate, its loading into ZIF-8 can be estimated from its size. For small adsorbates (e.g., water or methanol), the number of molecules within the pores is larger than for bulkier adsorbates (e.g., SFU or CAF), so the total U of the former systems, which are given by the intermolecular potential energies, are larger than that of the latter ones. As a result, the relative contributions of U_{HA} to U should increase with the size of the adsorbate. Indeed, that is the observation in Figure 2b, where the relative contribution of U_{HA} to U decrease as $CAF < SFU < \text{methanol} < \text{water}$ according to the decrease of the size of the adsorbate. The simulated systems are far from the saturation conditions (no high loadings were considered). These ensure that the adsorption enthalpies H_{ads} in Figure 2c and Table S3 calculated from the energy/particle fluctuations in the grand-canonical ensemble can be considered accurate.⁵⁰ For the pure drugs, the loadings are high, so not much emphasis should be placed on the H_{ads} of SFU being more negative than that of CAF. In fact, the difference in H_{ads} might also be due to an unbalance in the potential energy function of SFU.

A more concerning issue is that these GCMC simulations with the ZIF-8P model enforce the adsorption of SFU and CAF into the pores of ZIF-8, which is not representative of the actual adsorption process because it has been shown¹⁴ that the inner porosity of ZIF-8 is not accessible to large molecules. In addition, the adsorption occurs from a solution of the drug, so the solvent effects as well as the surface and the inaccessibility to inner pores should be considered in these simulations. Therefore, the adsorption from a methanol solution of the drugs into ZIF-8P model will be presented next and should provide information regarding the solvent effects. After, the

surface effects will be addressed by the ZIF-8S model, considering the inner pores inaccessible.

Adsorption of Drugs from Solution. Figure 3 presents the loadings of SFU and CAF into ZIF-8P from methanol

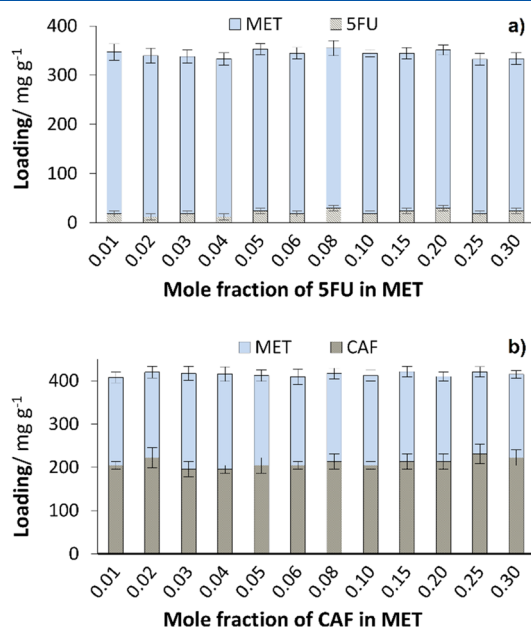


Figure 3. Adsorption into ZIF-8P from methanol solutions at different mole fractions represented by the loadings (mg g^{-1}) of (a) SFU and (b) CAF, with the respective methanol (MET) adsorption. Table S5 presents the results of these simulations.

solutions at mole fractions ranging from 0.01 to 0.30 obtained by GEMC simulations (see Table S4 for specific composition of the S_1 box). The methanol loadings from competitive adsorption with the drugs are also included in Figure 3.

For both drugs, their calculated loadings (mg g^{-1}) are almost constant in this mole fraction range and the total uptake (drug + solvent) from the solution is also nearly the same, ca. 350 and 400 mg g^{-1} for SFU and CAF, respectively. Consequently, the other quantities, such as U , U_{HA} , relative U_{HA} to U , and H_{ads} , are practically independent of the concentration in the range from 0.01 to 0.30 mole fractions, so they are presented in Figure 4 as averages over the concentrations with very small deviations.

Because of the competition with the solvent, the loadings of both drugs (Figure 4a) were smaller than those obtained for the adsorption of the pure substances. However, the adsorption of SFU in ZIF-8P (20 mg g^{-1}) decreases drastically (more than 20 times) when compared to the adsorption (477 mg g^{-1}) of the pure drug, whereas for CAF this decrease was only half. In fact, the loading of CAF (210 mg g^{-1} , ca. 17 wt %) is a better description of the experimental adsorption of this drug from methanol solution (12.2 wt %).¹⁴

An important observation is that when the adsorbed phases in S_0 achieve equilibrium with the bulk solution in S_1 , the values of U_{HA} in S_0 are almost the same as those for the simulations of pure drugs, namely, averages of -3.1×10^3 and $-3.5 \times 10^3 \text{ kJ mol}^{-1}$ (Table S5) for the SFU and CAF in methanol solutions, respectively. This indicates that the solvent (methanol) practically does not affect the interactions of the drugs with the ZIF-8P pores. However, because of the new drug–methanol interactions, U become more negative (-5.4

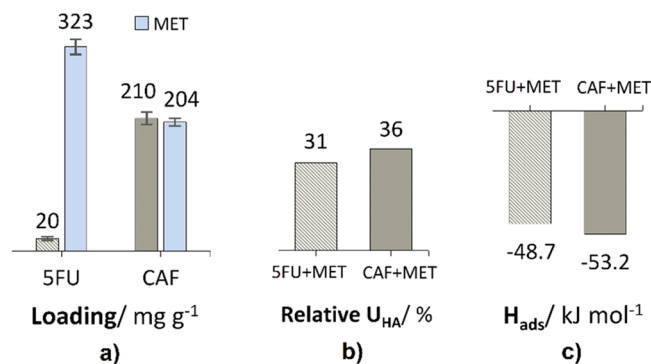


Figure 4. (a) Average loadings (mg g^{-1}) of the drugs and of methanol into ZIF-8P from methanol solutions at mole fractions ranging from 0.01 to 0.30 by GEMC simulations. (b) Average of the relative percentage of U_{HA} to U , expressed as $100 \times U_{\text{HA}}/U$. (c) Average of the adsorption enthalpy H_{ads} (kJ mol^{-1}) of drug + methanol. SFU = 5-fluorouracil, CAF = caffeine, MET = methanol, U_{HA} = host–adsorbate interaction energy, U = total internal energy.

$\times 10^3$ to $-9.8 \times 10^3 \text{ kJ mol}^{-1}$ for SFU and -5.1×10^3 to $-9.7 \times 10^3 \text{ kJ mol}^{-1}$ for CAF) and almost the same for both drugs (Tables S3 and S5). Consequently, the same U_{HA} now represent 31% and 36% of U in the adsorbed phases with SFU and CAF (Figure 4b), respectively. That is, the major contribution to the total energy comes now from adsorbate–adsorbate interactions and not from host–adsorbate as was for the pure-drug experiments (Figure 2). Indeed, H_{ads} for the methanol solutions (Figure 4c) are less negative than those for the pure drugs (Figure 2c), and they are closer to the values for pure methanol, which also makes the ZIF-8P structure favorable to load high amounts of the solvent. In this context, the low loading of SFU should be associated with stronger SFU–methanol interactions, responsible for the major contribution to the adsorbate–adsorbate energy (69% of U), and to stronger ZIF-8–methanol interactions, which is a major part of the 31% related to U_{HA} . Unfortunately, the individual contributions from SFU and methanol could not be separated with the current simulations. In fact, the preferential adsorption of methanol into ZIF-8P compared to SFU was captured by a short NVT-MD simulation of the adsorbed phase at the desired loading of 25 wt % (340 mg g^{-1}). It was observed that methanol molecules frequently displaced SFU molecules from their interaction sites on the framework, which was not observed for the case of CAF. It does appear that the interaction potentials between SFU–ZIF-8 and SFU–methanol should be reevaluated. However, before this reparametrization, the effects of the surface and accessibility to the inner pores should be considered.

Adsorption onto the Surface. As it has been pointed out, the diffusions of SFU and CAF through the pores of ZIF-8 are severely hindered by high energy barriers, so only the solvents (water and methanol) can access the pores of ZIF-8.¹⁴ In fact, these findings motivated to develop the present work and to propose alternative and hopefully more adequate procedures to model the adsorption of large drugs by ZIF-8. The ZIF-8S model was thus constructed (Figures 1, S3, and S4) and was employed in the simulations of the adsorption of drugs by the ZIF-8 surface by making the inner pores inaccessible to the drug but accessible to the solvent (methanol). Figure 5 shows the loadings, the relative U_{HA} , and the adsorption enthalpy H_{ads} obtained by GEMC simulations with the ZIF-8S model for the

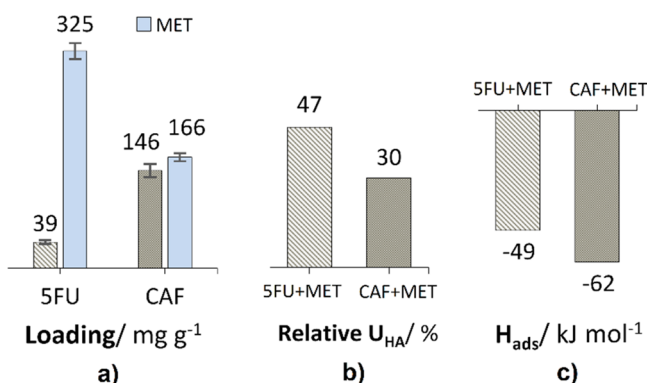


Figure 5. (a) Loadings (mg g^{-1}) onto ZIF-8S from methanol solutions (0.30 mole fraction) of the drugs by GEMC simulations. (b) Relative percentage of U_{HA} to U , expressed as $100 \times U_{\text{HA}}/U$. (c) Adsorption enthalpy H_{ads} (kJ mol^{-1}) of the drug + methanol. SFU = 5-fluorouracil, CAF = caffeine, MET = methanol, U_{HA} = host-adsorbate interaction energy, U = total internal energy.

adsorption from a 0.30 mole fraction methanol solution. The dependence with the concentration was not investigated because as shown for the ZIF-8P model, the adsorptions of these drugs were practically independent of the concentration, so the simulations were performed only for the largest mole fraction. Table S6 contains additional results from these simulations.

Despite the pores of the ZIF-8S model being inaccessible to SFU and CAF, but accessible to methanol, the loadings of the drugs were higher than the average loadings in the ZIF-8P model (Figures 4a and 5a). Loadings of $39 \pm 3 \text{ mg g}^{-1}$ (ca. 4 wt %) and $146 \pm 10 \text{ mg g}^{-1}$ (ca. 13 wt %) were obtained for SFU and CAF, respectively, which are in better agreement with experiments than those with the ZIF-8P model. For SFU, even with a 2-fold increase (20 to 39 mg g^{-1}) in the loading relative to the ZIF-8P model, the simulation still underestimates the experimental adsorption. Whereas, for CAF, the simulations with the ZIF-8 model coupled to the restricted access to the inner pores provided a remarkable agreement with the experimental result.

For this ZIF-8S model, the host-adsorbate (drug + methanol) energies U_{HA} are still the same ($-11.7 \times 10^3 \text{ kJ mol}^{-1}$ for SFU and $-11.3 \times 10^3 \text{ kJ mol}^{-1}$ for CAF); within the deviations, however, they have become much more negative than those obtained for the ZIF-8P model (Table S6). Whereas the relative contribution of U_{HA} to U increased from 31% to 47% for the SFU adsorption into ZIF-8P compared to

the ZIF-8S model, while for CAF this percentage decreased from 37% to 30%. (Figures 4b and 5b). These differences are to be expected because there are significant changes in the host–drug interactions for the ZIF-8P and ZIF-8S models. In the former, these interactions are dominated by the imidazolate moiety of the sorbent, whereas for the ZIF-8S model, the interactions with the coordination unsaturated zinc ions are dominant. In fact, these observations can be rationalized by the radial distribution functions (RDF) for some relevant atom pairs of SFU in the adsorbed phase as shown in Figure 6b.

The height and narrowness of the first peak in the ZIF-8S surface-oxygen RDFs with methanol and SFU suggest strong interactions between these sites (Figure 6a and 6b, respectively), once the inner pores are not available to the drugs. These interactions are centered in, but not limited to, the undercoordinated Zn ions (labeled as Zn2 in this model) on the ZIF-8S surface (Figure S4). Despite these strong interactions between the ZIF-8S surface and the SFU molecule, the adsorption of SFU from the methanol solution has been underestimated. This suggests that the SFU–methanol interactions are too strong (Figure 6c), so the transfer of SFU from the methanol solution to the surface is unfavorable and/or the ZIF-8S–methanol interactions are too strong such that the methanol molecules block the adsorption of SFU. Indeed, preliminary simulations with weaker ZIF-8S–methanol interactions provided higher loadings of the SFU onto the surface of ZIF-8. The refinement of interaction potentials is not a concern of the present work, however, considering the ZIF-8S model as the most representative of ZIF-8, the loadings onto this model could be used as criterion to refine and validate interaction potentials for adsorption of large drugs. This is one of the perspectives of the present work.

In summary, representing ZIF-8 by finite crystals with defined surfaces and blocking the inner pores access to the drugs is appropriate to describe the adsorption of CAF and SFU from a methanol solution, and can be used to refine or validate sorbent–sorbate interaction potentials. In addition, this model can explain the incorporation of other larger drugs by ZIF-8 once it is a surface process.

Desorption of the Drugs to Water. The adsorption step is certainly relevant in sorption applications, however, the desorption is essential for controlled drug release. Thus, both ZIF-8P and ZIF-8S models were applied in the simulations of the desorption of SFU and CAF to water. The initial conditions of these desorption simulations consisted of employing an equilibrated configuration obtained from the

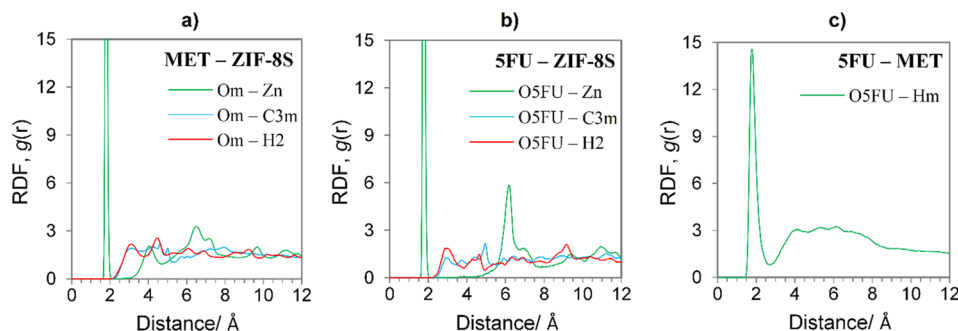


Figure 6. Radial distribution functions (RDF) of relevant atom pairs for (a) methanol–ZIF-8S, (b) SFU–ZIF-8S, and (c) SFU–methanol (MET) interactions in the adsorbed phase. Results correspond to the Zn, H2, and C3m centers of ZIF-8S (see Figure S1 for atom labels) with the oxygen centers of SFU (O5FU) and methanol (Om).

simulation of the adsorption of the drugs, with the solvent (methanol) molecules removed. This should simulate the experimental conditions, which dried the systems before performing the desorption experiment. As a result, there are three models for the desorption, namely, pZIF-8P that employed adsorption from the pure drugs into the ZIF-8P model, sZIF-8P, and sZIF-8S that started with the loadings obtained by GEMC simulations of the adsorption from a methanol solution. The results are summarized in Table S7, and Figure 7 presents the amount of drug released to water at

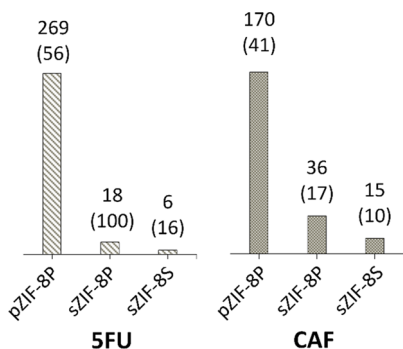


Figure 7. Release of 5FU and CAF from ZIF-8 models simulated by GEMC. Results are in milligrams per gram and percent (in parentheses) related to the loadings obtained from the following adsorption simulations: pure drug into ZIF-8P, drug from methanol solution into ZIF-8P, and drug from methanol solution onto ZIF-8S, labeled as pZIF-8P, sZIF-8P, and sZIF-8S, respectively.

room temperature. For all models, the desorption was larger for 5FU than CAF in terms of percentage released. In terms of milligram of released drug per gram of sorbent (mg g^{-1}), CAF is larger than 5FU for the sZIF-8P and sZIF-8S models because CAF has higher loadings than 5FU. As expected, the results obtained with the pZIF-8P model greatly overestimate the desorption, because the chemical potentials of the pure drugs within the pores of ZIF-8 are much larger than those in pure water. Considering the desorption of the drugs with more realistic loadings, the release (mg g^{-1}) decreased drastically. The 18 mg g^{-1} release of 5FU from the sZIF-8P model represents a complete desorption of the drug due to the unrealistic low loading (only three 5FU) of these molecules into ZIF-8P from the methanol solution.

These simulations yielded trends for the desorption that agrees with experiments, which show that lower loadings lead to larger and rapid releases.⁹ Quantitatively, however, the drug release estimates with the ZIF-8S model (columns sZIF-8S in Figure 7) are in much closer agreement with the experimental measurements, namely, 5.7% for 5FU and 12.5% for CAF after 5 min.¹⁴ The overestimated result for 5FU could be remediated by improving the sorbent–5FU interaction potential.

CONCLUDING REMARKS

The adsorption and release of two target drugs (5-fluorouracil (5FU) and caffeine (CAF)) into and from the ZIF-8 framework were simulated by the Gibbs-ensemble Monte Carlo approach employing two models representing the sorbent. One model without surface (ZIF-8P) and other with surface (ZIF-8S), for which the inner pores of ZIF-8 were inaccessible to the drugs, but accessible to the solvents (methanol or water). The adsorptions of both drugs into the

ZIF-8P model from a reservoir with the pure substances were greatly overestimated, which led to an overestimation of the amounts of drug released into water. Thus, such a model is not recommended to describe the actual sorption processes because it lacks surface and solvent effects.

The simulations of the adsorption of both drugs from a methanol solution modeled by explicit solvent molecules are more realistic and yielded more reliable results. However, the ZIF-8P model does not properly describe the sorption processes from and to a solution, because the drugs cannot diffuse through the pores of ZIF-8. In addition, this model has very specific and restrict sorbent–drug interactions that are dominated by a few interaction sites of the imidazolate organic linkers. These shortcomings are reflected in the poor quantitative agreements with the experimental results.

The ZIF-8S model yielded results for the sorption of CAF in very close agreement with the experimental loading from methanol solution and release of the drug into water. For 5FU, the computer simulations provided qualitative agreements, however, the adsorption was underestimated, which led to an overestimation of the percentage amount of 5FU released into water. This suggests that the sorbent–5FU interaction potentials should be improved. In fact, we propose to employ this ZIF-8S model and the experimental results regarding the sorption of these drugs to validate the interaction potentials for sorption processes. The excellent performance of the ZIF-8S model is due to its adequate description of the surface and by exposing adsorption sites such as undercoordinated zinc ions to interactions with large molecules. This was achieved by changing the periodic conditions from the atomistic level and applying this condition to a higher molecular level, for instance, a ZIF-8 nanocrystal. There are, of course, other techniques and methods to model surfaces of MOFs; however, the implementation of the proposed approach is quite simple and it can be generalized to describe several surface defects such as ionization, fragment rearrangements, or complete pore vacancies. Furthermore, the combination of this ZIF-8S model with the Monte Carlo method provide a very simple and efficient approach to simulate the inaccessibility of large molecules to the inner pores of ZIF-8. Namely, any trial moves that inserted the drug within the pore were disregarded. This is a quite simple and general approach that can be promptly applied to a large number of MOF sorbents and of drugs that cannot access the inner pores.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jcim.9b00893>.

Complementary figures, tables, and relevant pictures (PDF)

ZIF-8P model (CIF)

ZIF-8S model (CIF)

ZIF-8S.block file with the data for blocking the inner porosity of the ZIF-8S model (TXT)

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Notes

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ABBREVIATIONS

5FU, 5-fluorouracil; CAF, caffeine; GCMC, grand-canonical Monte Carlo; GEMC, Gibbs-ensemble Monte Carlo; H_{ads} , enthalpy of adsorption; L-J, Lennard-Jones; MC, Monte Carlo; MD, molecular dynamics; MET, methanol; MOF, metal-organic framework; PSD, pore size distribution; PXRD, powder X-ray diffraction; RASPA, the computer simulation package; RDF, radial distribution function; S_0 , simulation box for the adsorbed phase; S_1 , simulation box for the bulk liquid phase; U, internal energy; U_{HA} , host-adsorbate energy; VLE, vapor-liquid equilibrium; WAT, water; ZIF, zeolitic-imidazole framework; ZIF-8P, model of ZIF-8 without surface; ZIF-8S, model of ZIF-8 with surface

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